

Arnav Sagar

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Education

- **Delhi Technological University (DTU), Delhi** New Delhi, India
B.Tech. – Software Engineering; CGPA: 8.75 Jul 2024 – Jun 2028
- **Kendriya Vidyalaya Sector-47, Chandigarh** Chandigarh, India
12th: 93% 10th: 94.4% 2020 – 2023

Work Experience

- **Regavis Labs Pvt Ltd** New Delhi, India
Machine Learning Intern [Certificate] Jun 2026 – Present
 - Architected a scalable two-stage cascade architecture for real-time audio deepfake detection, integrating VAD, lightweight LFCC-LCNN screening, and SSL-based verification to reduce average compute cost by **85%** for live calling systems.
- **5G Lab (Department of Telecommunications, GoI), Delhi Technological University** New Delhi, India
Research Intern (5G, MEC, URLLC) [GitHub] [Certificate] Jun 2025
 - Engineered a **sub-25ms P95** on-device inference pipeline via per-channel entropy calibration, achieving **4x model compression** and **1.8x throughput** at <1.5% mAP loss, deployed across **CUDA, Metal MPS, and CPU** backends.
 - Built a real-time multi-person tracker using Lucas-Kanade optical flow with IoU-based keypoint association, feeding a **self-supervised trajectory autoencoder** (~1,600 params, <0.1ms/track) over a simulated **5G URLLC transport layer** (2ms latency, 0.1% packet loss). Presented at **PEC Chandigarh**.
- **AIMS-DTU** New Delhi, India
Research Intern (LLM Safety) [GitHub] [API] [Certificate] Jun 2025 – Jul 2025
 - Deployed a *session-aware* jailbreak & toxicity classifier with MiniLM cosine similarity, **SHAP** token attribution, adversarial normalization (homoglyph, zero-width, base64), **Redis** caching and an **ONNX**-optimized inference.

Relevant Projects

- **CLASP — Claude API Switching Proxy** [GitHub]
 - Architected a rate-limit-aware multi-provider proxy routing Claude Code/Codex traffic across 18 LLM inference providers using a pre-emptive Token Bucket rate limiter and multi-key pool rotation with circuit breakers to eliminate **429 errors**.
 - Engineered an async **priority queue with SSE keep-alive absorption** to buffer requests during quota exhaustion, plus a three-tier LRU/SQLite/semantic (FAISS) cache and an Anthropic-OpenAI protocol translator.
- **CAF-OTSRNet — Thermal Super-Resolution Model** [GitHub] [Demo]
 - Developed a **triple encoder cross-attention** fusion architecture for thermal super-resolution with two-stage optical-thermal alignment module and a **progressive Laplacian decoder** for multi scale reconstruction.
 - Integrated texture guided safety constraints and **physics aware losses** to suppress hallucinations, improving PSNR by **16%** and SSIM by **8.22%** compared to *SOTA model* as evaluated on the dataset provided by **ISRO**.
- **Retrieval Augmentation System** [GitHub]
 - Built a multi-stage RAG system processing 15k+ PDF pages with semantic chunking, hierarchical indexing, **HyDE-based query expansion**, and query decomposition, improving answer relevance by **38%** (measured via GPT-4 judge).
 - Combined hybrid dense-sparse retrieval (**FAISS + BM25**) with RRF fusion, cross-encoder reranking, **MMR** diversity selection, contextual compression, and **CRAG**-based hallucination suppression, reducing hallucination rate by **47%**.
- **Hybrid CTC-Attention Scene Text Recognition** [GitHub]
 - Designed a **CRNN** architecture that features a VGG-style CNN backbone, combining a 2-layer **BiLSTM** encoder with *Bahdanau attention* decoder for sequence modeling.
 - Implemented a joint CTC and Cross-Entropy loss with **beam search** to stabilize early training and refine character alignment, achieving **89%** word accuracy on the **IIIT5K** benchmark after training on **9M** images.
- **Text Splitting and Embedding Visualizer** [GitHub] [Demo]
 - Created an interactive visualization tool for analyzing text chunking strategies used in RAG pipelines and embeddings with Cosine semantic similarity matrix, recursive splitting, metrics and overlaps.
- **AI vs Human Text Classification System** [GitHub]
 - Benchmarked *8+ classical and ensemble models* (LR, SVM, Naive Bayes, Decision Trees, RF + AdaBoost, kNN, ANN and attention based **LSTMs**) against a **fine-tuned RoBERTa**, achieving an accuracy of **0.9996**.

Technical Skills

- **Languages:** Python, C/C++, JavaScript, HTML/CSS, SQL
- **ML/AI Frameworks:** PyTorch, TensorFlow, ONNX, HuggingFace, Transformers, OpenCV, Pandas, NumPy, LlamaIndex
- **Developer Tools & Backend:** REST APIs, AsyncIO, Redis, WebSockets, Docker, FastAPI, Weights & Biases, Git, Streamlit

Achievements

- National Finalist, Smart India Hackathon 2025 Ahmedabad, India

Positions of Responsibility

- Coordinator, Business Bulls, DTU (Finance & Strategy Club)